

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0601

Hewlett 100 kV
October 27, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0601. The Interconnection Customer proposes a 268.5 MW natural gas combined cycle facility interconnecting at the Hewlett 100 kV substation in the Illinois area. The request was received on March 17, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 250 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$432,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Comed / Somerset Operating Co Llc 100kV #1
Contingency: Comed / Somerset Operating Co Llc 100kV Line 2
Loading: 108.1% (Rating: 146 MVA)
DFAX: 9.9%
Pre-project loading: 89.9%
2. Facility: Comed / Somerset Operating Co Llc 100kV Cir. 2
Contingency: Comed / Somerset Operating Co Llc 100kV #1
Loading: 108.1% (Rating: 146 MVA)
DFAX: 9.9%
Pre-project loading: 89.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Comed / Somerset Operating Co Llc 100kV #1	Reconductor line	\$684,000	42% (\$290,000)
Comed / Somerset Operating Co Llc 100kV Cir. 2	Reconductor line	\$335,000	42% (\$142,000)
	Total allocated to project		\$432,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.